

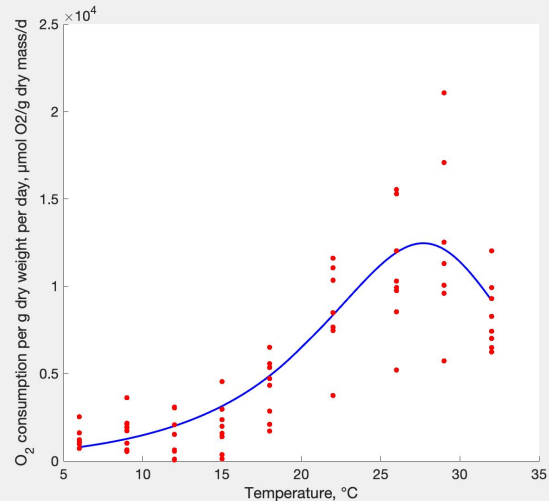
Symbiotic model - Updated 11.12.25

Astrangia poculata sym

COMPLETE = 3.0; MRE = 0.077; SMAE = 0.108; SMSE = 0.040

Data & predictions

data	prd	RE	symbol	units	description
1096	1099	0.00319	tp	d	time since birth at puberty
0.011	0.01099	0.0009003	L0	cm	egg diameter
0.0137	0.0137	0.000316	Lb	cm	total length at birth
0.8	0.7619	0.04762	RQ	-	Respiratory quotient
see figure		0.3077	TJO	Temperature	vs. O ₂ consumption per g dry weight per day
0.02	0.02372	0.1859	psd.v	cm/d	energy conductance
18	98.1	4.45	psd.p_M	J/d.cm ³	vol-spec som maint
0.002	0.002	0	psd.k_J	1/d	maturity maint rate coefficient
0.3	0.005328	0.9822	psd.k	-	maintenance ratio
0.8	0.8764	0.09546	psd.kap	-	allocation fraction to soma
0.8	0.8006	0.0007266	psd.kap_G	-	growth efficiency
0.95	0.95	0	psd.kap_R	-	reproduction efficiency



std parameters at 20.0°C

symbol	units	value	free	description
T_ref	K	293.1	0	Reference temperature
T_A	K	1.23e+04	1	Arrhenius temperature
z	-	0.0922	1	zoom factor
F_m	l/d.cm ²	6.5	0	{F_m}, max spec searching rate
kap_X	-	0.8	0	digestion efficiency of food to reserve
kap_P	-	0.1	0	faecation efficiency of food to faeces
v	cm/d	0.02372	1	energy conductance
kap	-	0.8764	1	allocation fraction to soma
kap_R	-	0.95	0	reproduction efficiency
p_M	J/d.cm ³	98.1	1	{p_M}, vol-spec somatic maint
p_T	J/d.cm ²	0	0	{p_T}, surf-spec somatic maint
k_J	1/d	0.002	0	maturity maintenance rate coefficient
E_G	J/cm ³	261.3	1	{E_G}, spec cost for structure
E_Hb	J	7.376e-11	1	maturity at birth
E_Hp	J	2.713	1	maturity at puberty
h_a	1/d ²	1.207e-15	1	Weibull aging acceleration
s_G	-	0.0001	0	Gompertz stress coefficient
T_AH	K	3.384e+04	1	Arrhenius temperature for upper boundary
T_H	K	302.4	0	upper boundary tolerance range
del_M	-	0.009189	1	shape coefficient for larvae
del_W	-	1	0	dry - ash free dry weight ratio
del_Y	-	0.01687	1	shape coefficient for egg
f	-	1	0	scaled functional response for 0-var data

Aposymbiotic model - Updated 11.12.25

Astrangia poculata apo

COMPLETE = 3.0; MRE = 0.097; SMAE = 0.147; SMSE = 0.041

Data & predictions

data	prd	RE	symbol	units	description
1096	1057	0.03561	tp	d	time since birth at puberty
0.011	0.01069	0.02847	L0	cm	egg diameter
0.0137	0.01398	0.02039	Lb	cm	total length at birth
0.8	0.7619	0.04762	RQ	-	Respiratory quotient
see figure		0.3153	TJO	Temperature	vs. O ₂ consumption per g dry weight per day
0.02	0.05092	1.546	psd.v	cm/d	energy conductance
18	148.7	7.259	psd.p_M	J/d.cm ³	vol-spec som maint
0.002	0.002	0	psd.k_J	1/d	maturity maint rate coefficient
0.3	0.003407	0.9886	psd.k	-	maintenance ratio
0.8	0.5499	0.3126	psd.kap	-	allocation fraction to soma
0.8	0.8262	0.03269	psd.kap_G	-	growth efficiency
0.95	0.95	0	psd.kap_R	-	reproduction efficiency

std parameters at 20.0°C

symbol	units	value	free	description
T_ref	K	293.1	0	Reference temperature
T_A	K	1.654e+04	1	Arrhenius temperature
z	-	0.03505	1	zoom factor
F_m	l/d.cm ²	6.5	0	{F_m}, max spec searching rate
kap_X	-	0.8	0	digestion efficiency of food to reserve
kap_P	-	0.1	0	faecation efficiency of food to faeces
v	cm/d	0.05092	1	energy conductance
kap	-	0.5499	1	allocation fraction to soma
kap_R	-	0.95	0	reproduction efficiency
p_M	J/d.cm ³	148.7	1	[p_M], vol-spec somatic maint
p_T	J/d.cm ²	0	0	{p_T}, surf-spec somatic maint
k_J	1/d	0.002	0	maturity maintenance rate coefficient
E_G	J/cm ³	253.2	1	[E_G], spec cost for structure
E_Hb	J	2.345e-10	1	maturity at birth
E_Hp	J	1.749	1	maturity at puberty
h_a	1/d ²	1.073e-15	1	Weibull aging acceleration
s_G	-	0.0001	0	Gompertz stress coefficient
T_AH	K	4.059e+04	1	Arrhenius temperature for upper boundary
T_H	K	300.4	0	upper boundary tolerance range
del_M	-	0.00745	1	shape coefficient for larvae
del_W	-	1	0	dry - ash free dry weight ratio
del_Y	-	0.01376	1	shape coefficient for egg
f	-	1	0	scaled functional response for 0-var data

